V. Evaluation

A. Evaluation Overview

This chapter evaluates the performance of the proposed Safe-Beam Assisted Power-Release Relay (SAPR-R) method, focusing on its ability to improve user satisfaction and service continuity under the aggregate EPFD constraint. The evaluation consists of three main parts. First, we observe the aggregate EPFD variation during the critical satellite pass over the GS, as well as the number of beams that must be shut down to satisfy the EPFD constraint. This analysis confirms the existence of EPFD violation and beam shutdown demand during the critical pass, and further supports the rationale of designing a relay-based recovery method. Second, we compare different interference mitigation methods in terms of service performance and relay capability. Finally, we further analyze the impact of different EPFD limits on user satisfaction to evaluate the stability and applicability of SAPR-R under different interference constraint conditions.

B. Simulation Setup

To evaluate the performance of the proposed SAPR-R method under the NGSO–GSO coexistence scenario, we design a simulation framework that combines STK and MATLAB. This framework is used to construct the LEO constellation, obtain the time-varying satellite geometry, and further evaluate the performance of different interference mitigation methods in terms of EPFD compliance, downlink capacity, and user satisfaction.

The simulation uses the following two main tools:

- **STK (Satellite Tool Kit):** STK is used to construct the OneWeb-like LEO constellation and generate the relative positions, visibility relationships, and beam coverage information among the LEO satellites, the GS, and users at each time slot. Based on the system information generated by STK, the dynamic geometric variation during the critical satellite pass near the GS can be simulated.
- **MATLAB:** MATLAB is used to process the system information generated by STK and perform offline preprocessing. This includes constructing the beam–user association, critical/helper satellite relationship, and critical-beam/helper-beam mappings. MATLAB is then used to execute the proposed SAPR-R method and calculate the EPFD and user satisfaction performance of each method at different time slots.

The main simulation parameters used in this chapter are summarized in Table I. The table integrates the settings related to the LEO constellation, GSO satellite, GS, LEO user ground station, and EPFD evaluation. These parameters are used in the following EPFD evaluation and user satisfaction comparison.

C. Compared Schemes

D. Evaluation Results

1) EPFD Compliance and Beam Shutdown:

2) User Satisfaction under Different User Loads:

3) Relay Capability and Satisfaction Distribution:

4) Impact of Different EPFD Limits:

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TABLE I
SIMULATION PARAMETERS.

LEO constellation	
Altitude (km)	1200
Orbit plane inclination	$87.9°$
Time slot	1 s
Maximum transmit power	16.8 W per satellite
Maximum antenna gain	34.33 dBi
Downlink frequency	11.7 GHz
Bandwidth	250 MHz
Antenna pattern	[1] $A \exp(\beta\phi)$
Maximum tilt angle	$10°$
Approximated antenna gain coefficient, (A)	4.0×10^3
Approximated antenna gain coefficient, (B)	-0.0671
GSO satellite	
Altitude (km)	35785.4
Longitude	30.6
Latitude	0
Downlink frequency	11.7 GHz
GSO ground station	
Minimum elevation angle	$10°$
Maximum receive gain	38.1 dBi
Noise temperature	240 K
Antenna pattern	[2]
LEO user ground station	
Maximum receive gain	40.95 dBi
Noise temperature	240 K
Antenna pattern	[2]
User demand	50 Mbps
EPFD	
EPFD limit (dB(W/m^2/1 MHz))	-173.4 [3]
Reference antenna diameter	60 cm